

# **QUICKLOAN MOBILE : ETHICAL DATA GOVERNANCE REVIEW**

## **Summary of Review Process**

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Client: QuickLoan Mobile | Role: Independent Data Governance Consultant |  
Jurisdiction: Republic of Ghana

*Legal Framework: Ghana Data Protection Act 2012 (Act 843)*

### **Deliverable 3: Summary of the Review Process**

#### **1. How Data Lifecycle and Classification Principles Were Used to Identify Risks**

The governance review of QuickLoan Mobile's data pipeline was structured around two complementary analytical frameworks: the Data Lifecycle Model and the Data Classification Framework. Applied together and mapped directly onto the QuickLoan data flow diagram, these tools transformed what might initially appear as a collection of isolated technical deficiencies into a coherent, stage-by-stage picture of systemic governance failure one that spans every phase of the data asset's life from collection to destruction.

The Data Lifecycle Model structured the review by decomposing the pipeline into its sequential governance stages: Collection, Storage, Processing, Use, Sharing, and Retention/Destruction. Beginning at the Collection stage, the Data Minimisation principle which requires that organisations collect only the personal data strictly necessary for the stated processing purpose immediately identified the most visible and egregious flaw in the pipeline: Step 1 of the diagram explicitly labels the User Mobile App as collecting "excessive PII: contact list, GPS, device logs." None of these data types have any legitimate nexus to micro-loan credit-risk assessment. Their collection violates Act 843, Section 17(1)(a), and introduces thousands of third-party contacts into QuickLoan's data estate without their knowledge or consent a secondary privacy violation that multiplies the company's legal liability.

Moving to the Storage stage, the diagram's label "no consent capture" on the flow from Step 2 (API Gateway) to Step 3 (Raw Data DB 'AllEvents') identified the central legal failure of the entire pipeline: there is no verified lawful basis for any data that reaches the Raw Data DB. Under Act 843, Section 18, explicit, informed, and freely given consent of the data subject is a mandatory prerequisite for lawful processing. Its complete absence means that every subsequent operation in the pipeline Preprocessing, ML inference, Customer Notifications, Analytics, and Third-Party Sharing is unlawful. This single failure at the Storage stage legally invalidates the entire downstream architecture.

The Data Classification Framework assigned all financial PII in the QuickLoan pipeline National ID numbers, mobile money account details, income data, loan history, contact lists, GPS data, and device logs to the Sensitive/Confidential tier: the highest protection level under Act 843, requiring the strictest access controls, the most rigorous handling procedures, and the shortest permissible retention periods. This classification immediately established a governance benchmark against which every

pipeline stage was evaluated. The result was a consistent finding of deficiency at each stage: the Raw Data DB holds Sensitive/Confidential PII without classification tags, access controls, or a retention policy; the Preprocessing Service processes raw PII without pseudonymisation or proxy variable governance; and the Analytics DB and Third-Party Partner receive identifiable financial PII without any masking, anonymisation, or formal Data Sharing Agreement.

At the Processing and Use stages, the Preprocessing Service's absence of pseudonymisation and audit logging was identified as a failure of data protection by design a core Act 843 principle because raw PII flows directly into the ML feature engineering process. Proxy variables including GPS location, contact list size, and device model were identified as the primary mechanism of potential indirect discrimination: these variables correlate with protected demographic characteristics (geographic origin, socioeconomic status, ethnicity) and introduce disparate impact into the LoanScore ML Model's outputs without the model explicitly encoding any discriminatory intent. The Decision Service, labelled "auto-decision / No transparency logging" in the diagram, was identified as the Use stage's critical failure: consequential credit decisions are made and executed with no audit trail, no explanation to the applicant, and no human rendering them both unaccountable and unchallengeable.

At the Sharing and Retention/Destruction stages, the Analytics DB's label "Stores PII without masking" applied to both instances visible in the diagram confirmed that Sensitive/Confidential PII is being retained in identifiable form for analytics purposes without any de-identification, and shared with a Third-Party Partner without a Data Sharing Agreement or any anonymisation process. The absence of any retention policy across the pipeline means QuickLoan holds PII indefinitely, creating an ever-expanding compliance liability and a permanently growing data breach surface. The Data Lifecycle Model made this structural continuity of governance failure immediately visible: a deficiency at the Collection stage cascades through every subsequent lifecycle stage, and the absence of governance controls at the Retention/Destruction stage means there is no defined endpoint to the risk.

## **2. How the Proposed Metric Ensures Ethical and Transparent Governance**

The reporting metric recommended in Deliverable 1 the Loan Approval Rate Disaggregated by Demographic Group, operationalised as the Disparate Impact Ratio (DIR) was selected because it directly addresses the most ethically consequential flaw identified in the QuickLoan diagram: the Decision Service operating with fully automated loan decisions and no transparency logging. An automated lending model making consequential credit decisions at scale without any mechanism for transparency, accountability, or fairness monitoring represents a fundamental ethical failure and in the context of Ghana's developing economy, where access to micro-credit is a meaningful driver of economic inclusion, it carries the very real risk of systematically excluding the most financially vulnerable citizens from access to credit precisely at the moment they need it most.

The DIR operationalises the principle of fairness-by-measurement. The metric is defined as the ratio of the ML model's loan approval rate for a given demographic subgroup disaggregated by gender, age band, geographic region, and income tier relative to the overall approval rate across all applicants. The four-fifths rule, the international standard adopted from the US Equal Employment Opportunity Commission and widely applied in algorithmic fairness auditing, establishes that a DIR

below **0.80** for any protected subgroup constitutes evidence of disparate impact and triggers a mandatory bias review. Monitored monthly via a Grouped Bar Chart (which displays approval rates side-by-side across demographic groups for instant cross-group comparison) and a Time-Series Line Chart (which plots each group's DIR trend against the 0.80 threshold reference line over time), the metric provides both a static snapshot of current fairness performance and a dynamic signal for detecting model drift the gradual degradation of fairness properties as the model encounters new applicant data distributions.

The DIR metric ensures ethical governance through three mechanisms. First, it provides early warning of discriminatory outcomes before they become systemic: by computing and reporting approval rate disparities monthly, the governance team detects emerging bias at the level of model outputs not merely inputs and responds with model retraining and independent fairness validation before a vulnerable demographic group experiences prolonged discriminatory denial of credit. Second, it creates an auditable evidence trail: every DIR computation, threshold breach alert, and bias remediation action is recorded, timestamped, and stored in the governance log. This documentation is precisely what Act 843 and the Ghana Data Protection Commission require as evidence of accountable, responsible automated decision-making. Third, it embeds ethical accountability into QuickLoan's operational rhythm rather than treating fairness as a one-time model audit: by making the DIR a standard monthly key performance indicator reported to the DPO, senior management, and in summary form to the DPC QuickLoan institutionalises the principle that fairness is an ongoing operational obligation, not a deployment-time checkbox.

Beyond legal compliance, the DIR metric serves a strategic commercial function: it provides QuickLoan with the evidence base to demonstrate, to regulators, investors, civil society, and users alike, that its automated lending decisions are fair, transparent, and aligned with its stated mission of financial inclusion. In Ghana's increasingly competitive and scrutinised fintech landscape, the ability to make this demonstration with auditable, quantitative evidence is a meaningful differentiator one that builds user trust, reduces regulatory risk, and positions QuickLoan as a responsible AI leader in West Africa's emerging digital financial services sector. The DIR metric, in this sense, does not merely measure ethical performance: it constitutes ethical performance, making QuickLoan's commitment to fairness visible, accountable, and real.

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